

```
#program:input and output
name=input("enter your name:")
age=int(input("enter your age"))
print("Hello",name)
print("you are",age,"years old")
[3]
```

Hello Ramya

you are 19 years old

```
#program:Assignment operator
```

```
x=10
```

```
print("Initial value:",x)
```

```
x+=5#x=x+5
```

```
print("After +=:",x)
```

```
x-=3#x=x-3
```

```
print("After -=:",x)
```

```
x*=2#x=x*2
```

```
print("After *:=:",x)
```

```
x/=4#x=x*2
```

```
print("After /:=:",x)
```

```
x//=2#floor division
```

```
print("After //:=:",x)
```

```
x%=3#modulus
```

```
print("After %:=:",x)
```

```
[20]
```

```
Initial value: 10
```

```
After +=: 15
```

```
After -=: 12
```

```
After *:=: 24
```

```
After /:=: 6.0
```

```
After //:=: 3.0
```

```
After %:=: 0.0
```

```
#program:Relational and logical operations
a=int(input("enter first number:"))
b=int(input("enter second number:"))
print((a>2)and(b>=6))
#logical AND
print(True and True)#True
print(True and False)#False
#logical Or
print(True or False)#True
#logical NOT
print(not True)#False
[10]
```

True

True

False

True

False

```
#program:Arithmetic operations
a=int(input("enter first number:"))
b=int(input("enter second number:"))
print('sum:',a+b)
print('difference:',a-b)
print('multiplication:',a*b)
print('division:',a/b)
print('Floor Division:',a//b)
print('Modulo:',a%b)
print('power:',a**b)
```

[4]

```
sum: 6
difference: 2
multiplication: 8
division: 2.0
Floor Division: 2
Modulo: 0
power: 16
```

```
#program:Identity operators
```

```
a=10
```

```
b=10
```

```
print("a is b:" ,a is b)
```

```
print("a is not b:" ,a is not b)
```

```
[24]
```

```
a is b: True
```

```
a is not b: False
```



```
#program:membership operators
numbers=[10,20,30,40,50]
#check if 20 is in the list
print("Is 20 is in the list?",20 in numbers)
#check if 60 is not in the list
print("Is 60 not in the list?",60 not in numbers)
```

```
[21]
```

```
Is 20 is in the list? True
```

```
Is 60 not in the list? True
```

```
#Different list objects  
x=[1,2,3]  
y=[1,2,3]  
print("x is y:",x is y)  
print("x==y:",x==y)  
[25]
```

x is y: False

x==y: True

```
#program:Largest of three numbers
a=int(input("Enter first number:"))
b=int(input("Enter second number:"))
c=int(input("Enter third number:"))
if a>b and a>c:
    print("Largest:",a)
elif b>c:
    print("largest:",b)
else:
    print("largest:",c)
```

[5]

largest: 12


```
#program to check even or odd  
num=int(input("enter a number:"))  
if num%2==0:  
    print("Even number")
```

[27]

Even number

```
1 #program:print 1 to 5 using for loop
2 ✓ for i in range(1,6):
3     print(i)
4
5
```

[8]

1

2

3

4

5

```
#program:positive,negative or zero
num=int(input("enter a number:"))
if num>=0:
    print("Non-negative number")
else:
    print("Negative number")

[1]
```

Negative number

```
#program:print 1 to 5 using while loop
```

```
i=1
```

```
while i<=5:
```

```
    print(i)
```

```
    i+=1
```

```
[6]
```

```
1
```

```
2
```

```
3
```

```
4
```

```
5
```

```
#program:Multiplication table  
num=int(input("enter a number:"))  
for i in range(1,11):  
    print(num,"x",i,"=",num*i)
```

[10]

2 x 1 = 2

2 x 2 = 4

2 x 3 = 6

2 x 4 = 8

2 x 5 = 10

2 x 6 = 12

2 x 7 = 14

2 x 8 = 16

2 x 9 = 18

2 x 10 = 20

```
#program:sum of first N numbers
n=int(input("enter a number:"))
sum=0
for i in range(1,n+1):
    sum+=i
    print("sum:",sum)
```

[9]

sum: 1

sum: 3

sum: 6

sum: 10

sum: 15

sum: 21


```
#program:Factorial using loop
num=int(input("enter a number:"))
fact=1
for i in range(1,num+1):
    fact*=i
    print("Factorial:",fact)
```

[12]

Factorial: 1

Factorial: 2

Factorial: 6

Factorial: 24

Factorial: 120

Factorial: 720

```
#program:count digits in a number
num=int(input("enter a number:"))
count=0
while num!=0:
    num=num//10
    count+=1
    print("Number of digits:",count)
```

[11]

Number of digits: 1

Number of digits: 2

Number of digits: 3

Number of digits: 4

Number of digits: 5

Number of digits: 6

Number of digits: 7

Number of digits: 8